



Lecture 9: Rocky Planets I

Earth & Venus

Planetary Systems

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Learning Objectives

By the end of this lecture, you will be able to:

- Summarise Earth's interior, surface, ocean, and climate as the reference planet
- Compare Venus to Earth across mass, radius, atmosphere, and rotation
- Explain why a saturated water atmosphere has a maximum thermal flux
- Derive the Simpson–Nakajima runaway greenhouse limit
- Describe the D/H smoking-gun evidence for past water on Venus
- Outline current and upcoming Venus missions and the questions they target

Recap: Lectures 1–8

- Solar system formed from a disk of gas and dust around 4.567 Ga
- Heat sources: accretion, radiogenic, tidal, residual primordial
- Differentiation produces a layered interior: crust, mantle, core
- Atmospheric escape, magnetospheres, surface processes
- Earth's interior: $C/MR^2 = 0.331$, liquid outer core, vigorous convection

Today: the two innermost rocky planets, very similar in bulk parameters, very different in evolution. **Why?**

1. Earth as Reference

Earth seen from ~ 6 billion km. Voyager 1 (1990). Credit: NASA/JPL-Caltech

Earth at a Glance

Parameter	Value	Note
Mass	5.972×10^{24} kg	defines $M_{\oplus}$
Radius (mean)	6371 km	defines $R_{\oplus}$
Mean density	5515 kg m^{-3}	uncompressed $\sim 4030 \text{ kg m}^{-3}$
Semimajor axis	1.000 AU	defines AU
Bond albedo	0.30	cloud + ice + land
T_{eq}	255 K	equilibrium, no greenhouse
T_{surf}	288 K	+33 K greenhouse warming
Rotation period	23.93 h	spin axis tilt 23.4°
Magnetic field	dipolar, $\sim 25\text{--}65 \mu\text{T}$	active dynamo

Source: NASA Earth Fact Sheet (NSSDC)

What Makes Earth Habitable?

Four ingredients, all present, all stable for ~ 4 Gyr:

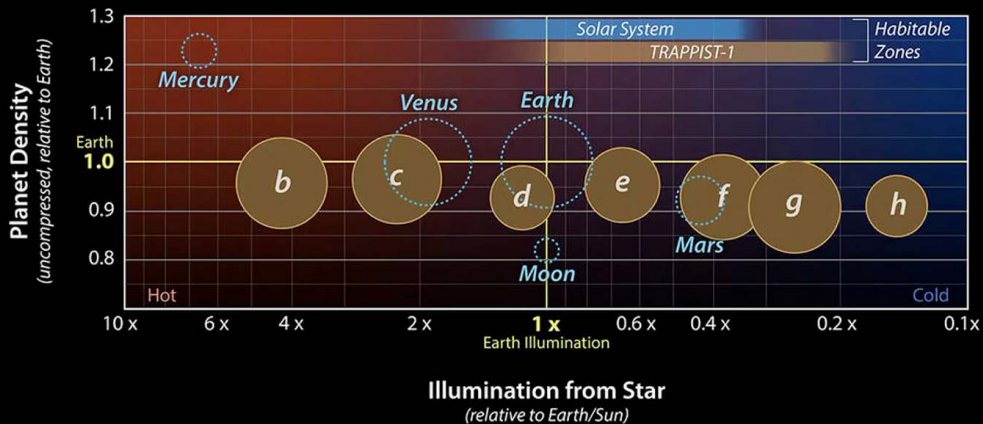
1. **Liquid water** on the surface, continuously
2. **Plate tectonics**: long-term carbon cycle, climate buffering
3. **Magnetic field**: shields atmosphere from solar wind erosion
4. **Stable orbit**: low eccentricity, large stabilising Moon

None of these is unique to Earth in principle.

All four together, on the same planet, at the same time, may be.

The puzzle is not that Earth is special. It is to identify which features are necessary, which are sufficient, and which are accidental.

TRAPPIST-1/Solar System Comparison



Credit: NASA

Why Compare to Other Systems?

Earth is one of ~ 5800 confirmed planets. The natural comparison set:

- Inner solar system: 4 rocky planets, 0.4–1.5 AU, around a G-type star
- TRAPPIST-1: 7 Earth-sized planets, 0.011–0.062 AU, around a late M dwarf
- Both systems span the inner edge of the habitable zone
- Comparable orbital architecture, drastically different host stars

Earth is the calibration point. Whether it is the typical or atypical outcome of rocky-planet evolution is exactly what we are trying to find out.

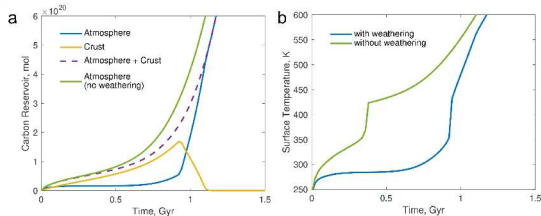
Plate Tectonics: The Engine

- Earth's lithosphere (rigid outer shell) is broken into ~ 15 major plates moving at cm yr^{-1}
- Three boundary types:
 - **Divergent** (mid-ocean ridges): new crust forms
 - **Convergent** (subduction zones, one plate sinks beneath another; mountain belts): crust destroyed or thickened
 - **Transform** (e.g. San Andreas): lateral slip
- Driven by mantle convection, slab pull, ridge push
- Recycles surface volatiles into the mantle on ~ 100 Myr timescales

No other body in the solar system has active plate tectonics today.

Preview: A Stagnant-Lid Reference Evolution

Reference scenario for
stagnant-lid (one immobile plate)
Venus over 4.5 Gyr:



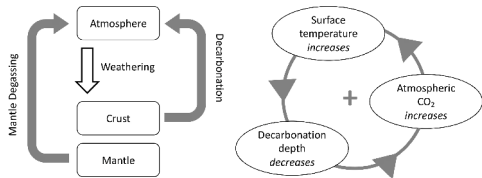
Credit: Höning et al. (2021)

- **Left:** carbon reservoirs (atmosphere vs. crust)
- **Right:** surface T with vs. without weathering
- Weathering keeps the surface temperate for ~ 0.9 Gyr
- Once water is lost, decarbonation drives runaway to a Venus-like end state

The Stagnant-Lid Carbon Cycle (Venus Preview)

Three reservoirs (mantle, crust, atmosphere) coupled by:

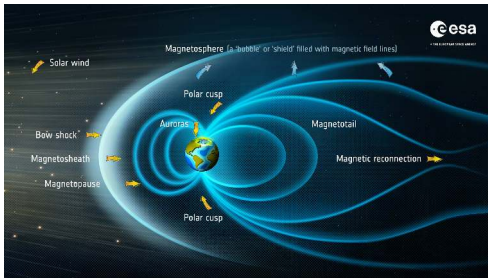
- **Mantle degassing** (volcanism): mantle to atmosphere
- **Surface weathering**: atmosphere to crust
- **Decarbonation**: crust back to atmosphere when buried carbonate heats up



Credit: Höning et al. (2021)

Destabilising loop on Venus: $T_s \uparrow \Rightarrow$ shallower decarbonation $\Rightarrow$ atmospheric $\text{CO}_2 \uparrow \Rightarrow T_s \uparrow$. No subduction returns carbon to the mantle.

Earth's Magnetic Field



Credit: ESA, CC BY-SA 3.0 IGO

- Generated by convection in the liquid outer core (Lecture 4)
- Mostly dipolar, surface intensity 25–65 μT
- Magnetopause stand-off: $\sim 10 R_{\oplus}$ on the dayside
- Reverses polarity on 10^5 – 10^6 yr timescales (paleomagnetism)
- Active for at least 3.45 Ga (Tarduno et al. 2010)

Key consequence: deflects solar wind, suppresses ion-pickup escape of O^+ , N_2^+ , H^+ .

Comparison to unmagnetised Mars (Lecture 10) shows the magnitude of this effect.

Hydrosphere and Cryosphere

Liquid water reservoirs:

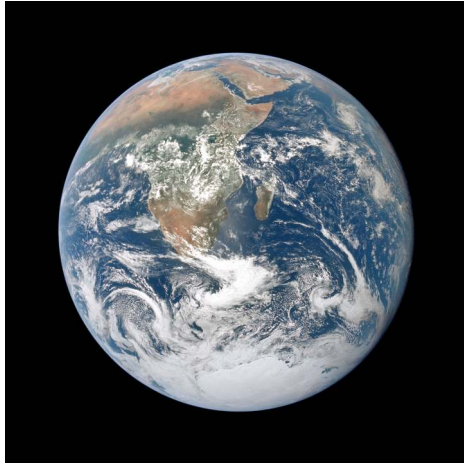
- Oceans: $1.34 \times 10^9 \text{ km}^3$
- Groundwater: $\sim 2.3 \times 10^7 \text{ km}^3$
- Lakes, rivers, soil moisture: small

Ice reservoirs:

- Antarctica: $26.5 \times 10^6 \text{ km}^3$
- Greenland: $2.85 \times 10^6 \text{ km}^3$
- Mountain glaciers, permafrost: smaller

- Total surface water $\approx$ 2700 m global equivalent layer (GEL)
- Ocean is the largest pool, but the deep mantle holds $\geq 1\text{--}2$ ocean masses
- Ice albedo provides a strong positive feedback in cold climate excursions

The Blue Marble



Credit: NASA / Apollo 17, image AS17-148-22727

Oceans (hydrosphere) and the sunlit Antarctic ice cap (cryosphere) in a single view.

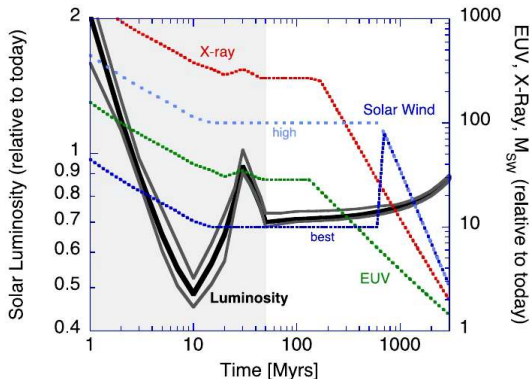
Climate Stability and the Faint Young Sun

Earth's climate has been stable enough for liquid water for ~ 4 Gyr, but not constant.

- Solar luminosity at 4 Ga: ~ 75% of today
- Without a strong greenhouse, Earth would have been frozen
- Geological evidence (sedimentary rocks, evaporites): liquid water at ≥ 3.8 Ga
- Compensating mechanism: higher $\text{CO}_2 + \text{CH}_4$, dimmer surface, weaker weathering

The Faint Young Sun problem is solved by the carbon cycle's negative feedback, not by accidents of initial conditions.

Solar Luminosity Through Time

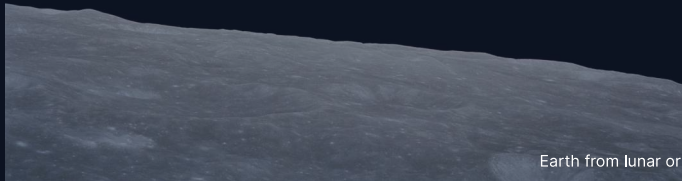


Credit: Zahnle et al. (2007)

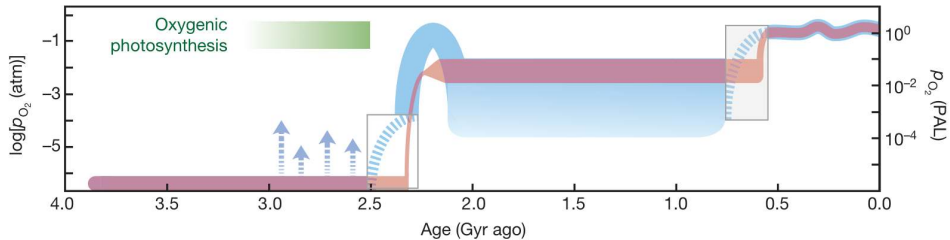
- Sun has brightened ~ 30% since formation
- Habitable-zone inner edge has migrated outward
- Venus inside it for at least the last ~ 1 Gyr
- The boundary swept past Venus' orbit in the past

Question for later: was Venus on the wet side when the boundary swept past?

2. Earth's Atmosphere and Biosphere



Earth from lunar orbit, Apollo 8 (1968). Credit: NASA



Reading the Oxygen History

Two great rises in atmospheric O_2 :

- **Great Oxidation Event** (GOE) at ~ 2.4 Ga: trace → ~ 1% of present atmospheric level (PAL)
- Long Mesoproterozoic plateau at 1%–10% PAL
- **Neoproterozoic Oxygenation Event** (NOE) at ~ 0.6 Ga: → near modern levels
- Phanerozoic fluctuations of ~ 15–35%

Drivers: cyanobacterial photosynthesis + organic carbon burial + decreasing reductant flux from the mantle.

Co-evolution of biosphere, atmosphere, and tectonics, across more than 2 Gyr.

Snowball Earth Episodes



Credit: Oleg Kuznetsov (3depix.com), CC BY-SA 4.0

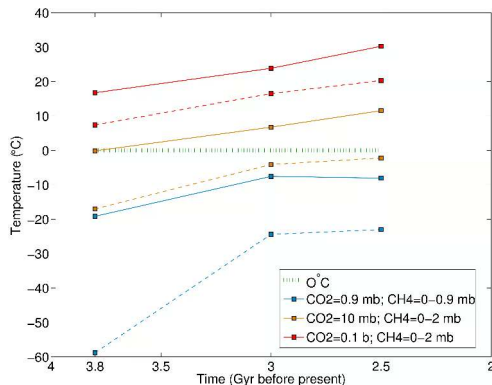
At least two global glaciation events in the Neoproterozoic:

- **Sturtian glaciation** (~ 717–660 Ma)
- **Marinoan glaciation** (~ 650–635 Ma)
- Glacial deposits at near-equatorial paleolatitudes
- Cap carbonates immediately above glacial diamictites

Mechanism: ice albedo runaway $\Rightarrow$ frozen oceans $\Rightarrow$ weathering halts $\Rightarrow$ volcanic CO_2 accumulates $\Rightarrow$ extreme greenhouse triggers deglaciation.

Source: Hoffman et al. (1998, 2017)

The Archean Climate Problem

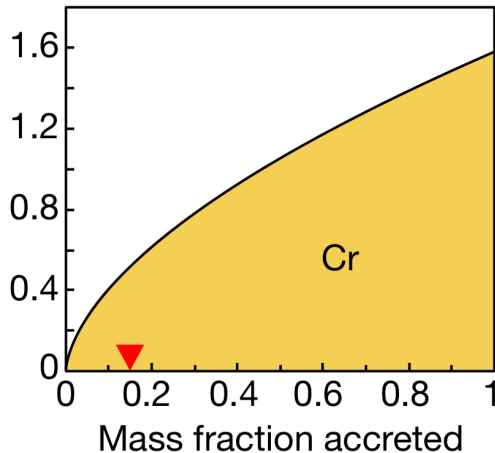


Credit: Charnay et al. (2013)

- Solar luminosity at 3 Ga: ~ 80% today
- Yet evidence for liquid water + biosphere
- Required greenhouse: CO₂, CH₄, possibly N₂ pressure broadening
- Constraints from paleosols, banded iron formations (ancient sediments)

Archean Earth was kept warm by an order of magnitude more greenhouse gas than today, not by a closer Sun.

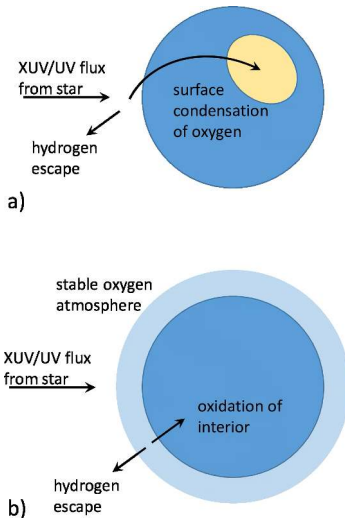
Earth's Bulk Composition: Late Accretion



Credit: Dauphas (2017)

- Bulk Earth matches non-carbonaceous (NC, inner-disk) material in $\epsilon^{54}\text{Cr}$ (panel shown; same story in O, Ti, Ni, Mo, Ru)
- Most of Earth's mass came from inner-disk material
- Late accretion ($\lesssim$ a few %): a small carbonaceous component, the "late veneer"
- Earth has a specific isotopic signature, not a generic chondrite mix

Abiotic Pathways to O_2

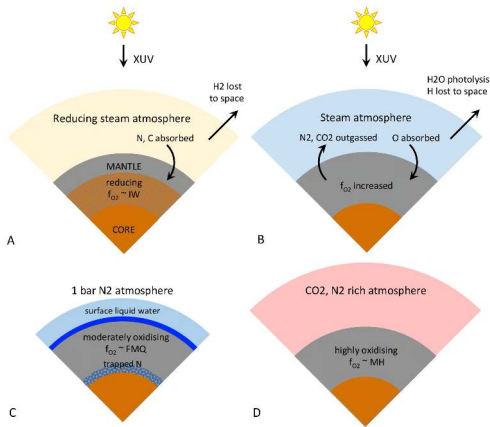


Credit: Wordsworth & Pierrehumbert (2014)

For exoplanet biosignatures: not all O_2 implies life.

- H_2O photolysis + H escape leaves abiotic O_2
- Especially at runaway-greenhouse conditions (surface too hot for liquid water)
- Or in CO_2 -rich atmospheres where $CO + O_2$ recombination is slow
- Earth's O_2 context-dependent biosignature

The N_2 Redox Pump



Credit: Wordsworth (2016)

- Atmospheric N_2 inventory is set by the surface + interior redox state
- **Earth path A→C:** surface ocean condenses; ends in 1-bar N_2 over a moderately oxidising mantle
- **Venus path A→B→D:** H escape oxidises mantle, ends in hot $CO_2 + N_2$ over a highly oxidised interior
- Abiotic redox pump: water-loss history drives the divergence, no biology required

The Biosphere as Geological Force

Life is not a passive passenger of Earth's climate; it is a geological agent.

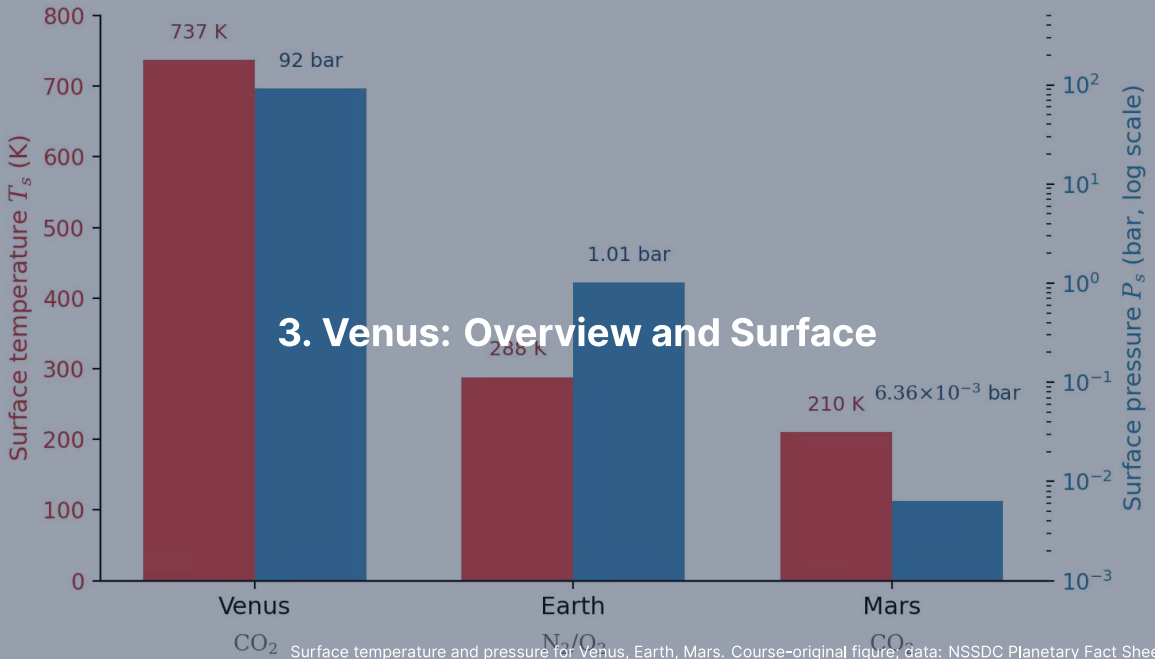
- **Banded iron formations** (~ 3–1.8 Ga): biological O_2 + dissolved Fe^{2+}
- **Continental crust composition**: shifted by oxidative weathering
- **Ocean redox state**: from anoxic to oxic, Snowball linked to NOE
- **Atmospheric N_2 inventory**: biological denitrification + nitrification

Earth's atmosphere is in extreme **thermodynamic disequilibrium** (O_2 + CH_4 coexist) maintained continuously by biology.

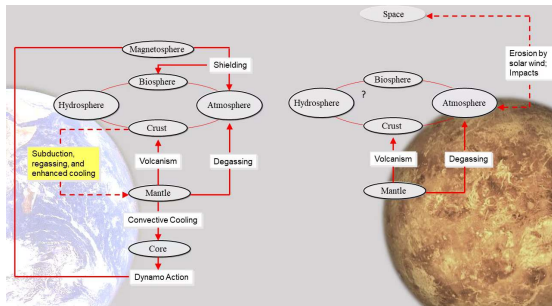
Earth as Reference: Summary

- Differentiated, magnetised, tectonically active rocky planet at 1 AU
- Liquid surface water, continuously, for ~ 4 Gyr
- Long-term carbon cycle buffers climate against luminosity changes
- Biosphere: a major geochemical reservoir, sink/source for many volatiles
- Plate tectonics + magnetic field + biosphere + ocean: the four-way coincidence

Now to the planet that started as Earth's apparent twin and ended as something completely different.



Earth and Venus as Coupled Systems



Credit: Gillmann et al. (2022)

Both are coupled systems of:

- Interior + tectonics
- Magnetosphere or its absence
- Atmosphere + clouds
- Surface + (possible) hydrosphere

The differences between them are emergent properties of which feedback loops close.

Venus and Earth: Twin and Not

Parameter	Earth	Venus
Mass ($M_{\oplus}$)	1.000	0.815
Radius ($R_{\oplus}$)	1.000	0.949
Mean density (kg m^{-3})	5515	5243
Semimajor axis (AU)	1.000	0.723
Stellar flux ($S_{\oplus}$)	1.00	1.91
Bond albedo	0.30	0.77
T_{surf} (K)	288	737
Surface pressure (bar)	1.0	92
Atmosphere	N_2/O_2	CO_2/N_2
Rotation period (d)	+1.0	-243 (retrograde)
Magnetic field	$\sim 50 \mu\text{T}$	$< 10^{-5} \times \text{Earth}$

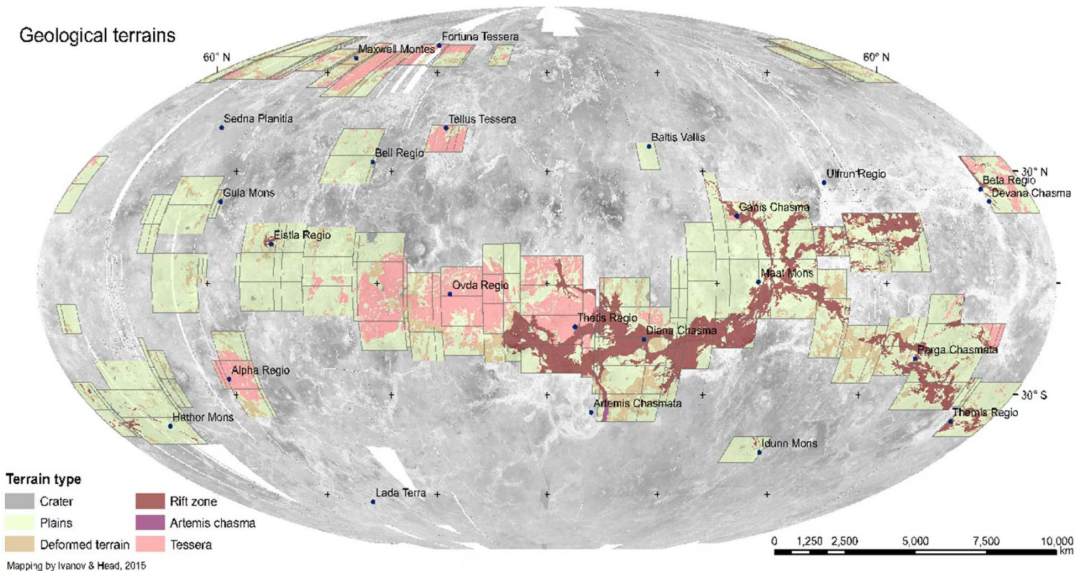
Source: NASA Venus Fact Sheet (NSSDC)

Venus: A Long Mission History

- **Mariner 2** (1962): first successful planetary flyby; thermal $T_{\text{surf}} \sim 700 \text{ K}$
- **Venera series** (1961–1984, USSR): 13 atmospheric probes, 8 surface landers
 - Venera 4: first *in situ* atmospheric composition
 - Venera 9: first surface images (1975)
 - Venera 13/14: surface composition (XRF), basaltic crust
- **Magellan** (1989–1994, NASA): radar mapping at $\sim 100 \text{ m}$ resolution, nearly global
- **Venus Express** (2005–2014, ESA): UV/IR atmospheric monitoring
- **Akatsuki** (2015–present, JAXA): atmospheric dynamics, super-rotation (clouds circling far faster than the planet)

No surface lander has survived more than ~ 2 hours: deep-ocean-trench pressures at red heat.

Geological terrains



Reading the Venus Surface Map

Categories highlighted in the Widemann et al. (2023) overlay:

- **Plains** (~ 80% of surface): basaltic, low elevation
- **Tessera**: highly deformed highlands, possibly oldest crust
- **Rift zones** (e.g. Beta Regio) and **deformed terrain**
- **Artemis chasma**: largest single tectonic feature
- **Craters** (~ 1000 catalogued, ≥ 2 km)

Coronae (circular tectonic-volcanic features) are a separate generic Venus class, not a category in this overlay. Volcanic edifices: > 85,000 catalogued globally.

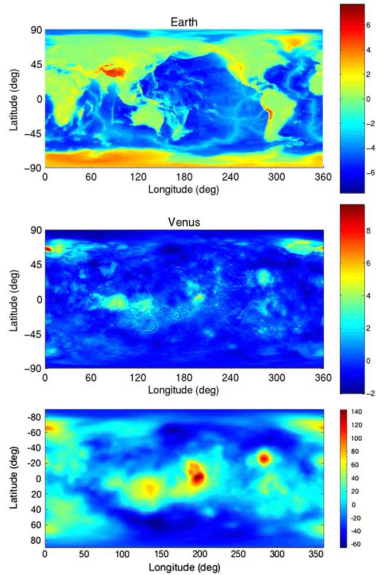
Key observation: **no spatial pattern of crustal age**. No oceanic-style mid-ocean ridges, no clear subduction belts.

Crater Statistics: A Single Surface Age

Venus crater statistics from Magellan:

- ~ 1000 identified impact craters, ≥ 2 km diameter
- Distribution is **statistically random**, no major terrain younger than another
- Mean crater density implies a global surface age of ~ 300–1000 Ma
- No areas of clearly archaic (> 1 Ga) terrain
- No active subduction belts (no oceanic-style recycling)

Venus' surface was either resurfaced globally and recently (~ 750 Ma catastrophic-resurfacing model) or is being resurfaced steadily (equilibrium model). Crater statistics alone cannot decide between them.

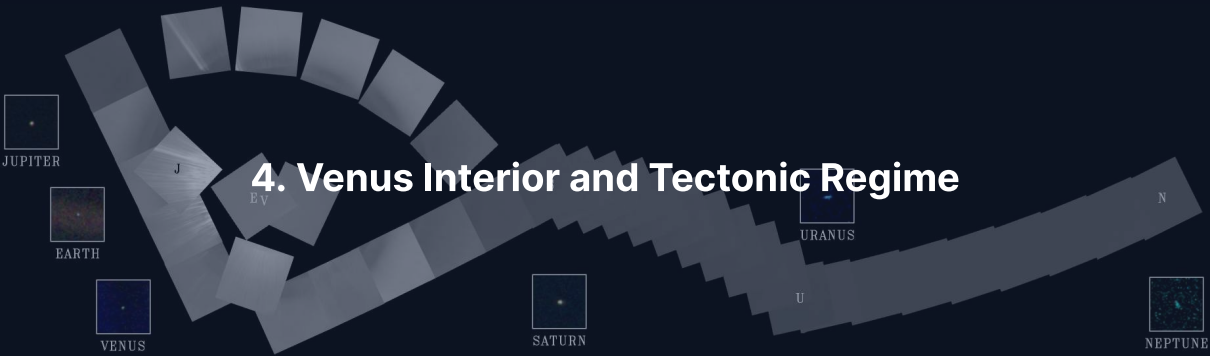


Earth topography, Venus topography, Venus geoid. Smrekar et al. (2018)

Why the Topography Differs

- Earth: **bimodal hypsometry** (distribution of surface elevations; oceans ~ -4 km vs. continents ~ +1 km)
- Venus: **unimodal hypsometry**, single dominant elevation
- Earth's bimodality is a direct signature of plate tectonics + isostasy
- Venus has highlands (Aphrodite, Ishtar) but no continents in the Earth sense
- Geoid/topography ratio: deep compensation depths on Venus

Hypsometry alone is enough to rule out Earth-style plate tectonics on Venus.



Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Venus' Interior: Inferred, Not Measured

No seismometer has ever operated on Venus.

Interior structure is inferred from bulk parameters and Earth-analog models.

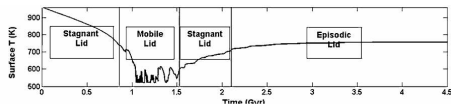
- Mean density 5243 kg m^{-3} ; uncompressed $\sim 4060 \text{ kg m}^{-3}$
- $C/MR^2 \approx 0.33$ (assumed Earth-like, not measured)
- Implied core radius: $R_c \approx 3000\text{--}3500 \text{ km}$ ($\sim 0.55 R_V$)
- Mantle composition: probably similar to Earth's
- Core state: possibly partially molten, but no strong evidence either way

Key observational gap: no C/MR^2 , no seismicity, no high-resolution gravity.

Venus Tectonic Evolution: A Model Trajectory

Surface temperature evolution from one numerical model: successive intervals of

- **Stagnant lid**
- **Mobile lid**
- **Stagnant lid**
- **Episodic lid**



Credit: Smrekar et al. (2018)

Conceptual alternatives debated for Venus today: **steady stagnant** (localised mantle plumes) vs. **episodic catastrophic overturn**. Either way, no Earth-style plate tectonics.

The Missing Magnetic Field

- Internal magnetic field $< 10^{-5}$ that of Earth (Pioneer Venus, 1978)
- Possible reasons:
 1. Core not convecting (insufficient heat flux)
 2. Core fully solid (already crystallised)
 3. Core fully liquid but subadiabatic (temperature gradient too shallow to drive convection)
 4. Slow rotation reduces dynamo efficiency
- Most likely: **stagnant lid** suppresses core heat loss $\Rightarrow$ no convection in core

Without a magnetosphere, the upper atmosphere is directly exposed to the solar wind.

This is a major channel for Venus' water loss.

Why Stagnant Lid?

Earth-style plate tectonics needs:

1. A weak, brittle, water-saturated lithosphere
2. Negative thermal buoyancy at subduction zones
3. Continuous mantle return flow

Venus likely fails on point 1: hot, dry surface $\Rightarrow$ strong, ductile lithosphere.

Coupled-feedback view: no liquid water $\Rightarrow$ dry strong lithosphere $\Rightarrow$ no plate motion $\Rightarrow$ no carbon-cycle return $\Rightarrow$ no water (one-way reinforcing loop).

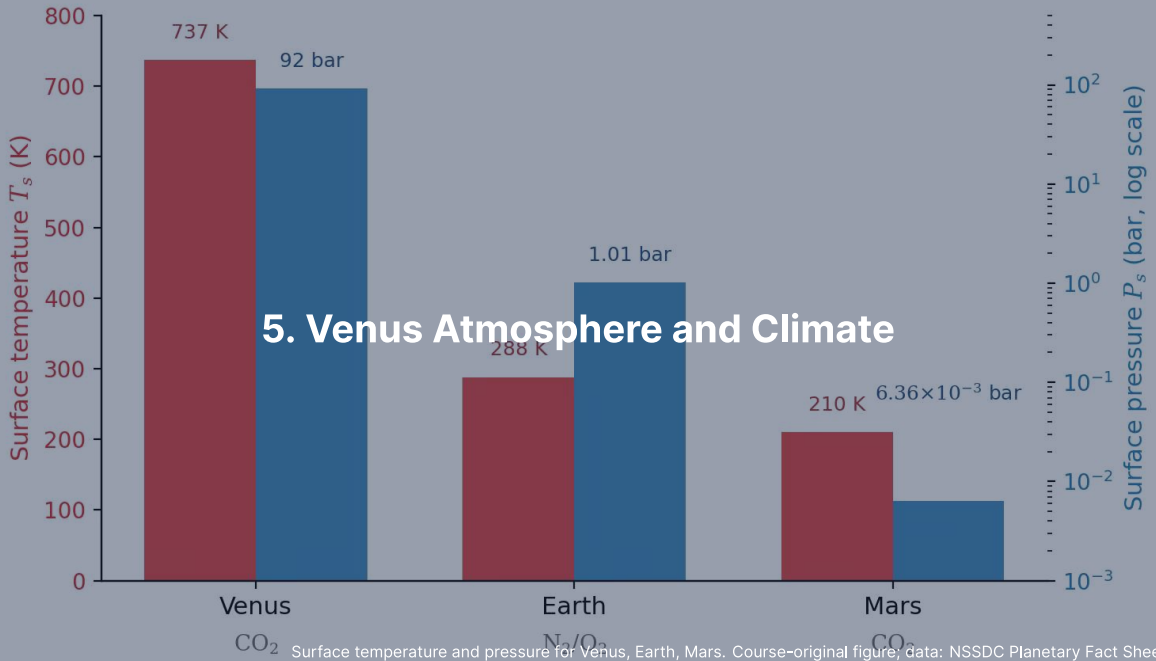
Source: Smrekar et al. (2018); Gillmann et al. (2022)

Is Venus Volcanically Active Today?

Three independent pieces of evidence for recent or ongoing volcanism:

- **Magellan radar reanalysis** (Herrick & Hensley 2023): a $\sim 2 \text{ km}^2$ vent on Maat Mons changed shape between 1991 and 1992
- **Venus Express thermal emissivity**: anomalously high emission from Idunn Mons, consistent with $< 250 \text{ kyr}$ lava flows
- **Atmospheric SO_2 variability**: order-of-magnitude changes on $\sim$ decadal timescales, consistent with episodic outgassing

Venus is almost certainly volcanically active today. The questions are: at what rate, and from how many vents?



Venus Atmosphere: Structure

- Surface: 737 K, 92 bar, 96.5% CO₂, 3.5% N₂
- Sub-cloud haze: 30–48 km
- **Main cloud deck:** 48–70 km, H₂SO₄ droplets, ~ 75% wt
- Mesosphere + thermosphere above, with a hot dayside / cold nightside contrast
- No tropopause cold trap of the Earth kind
- Lower atmosphere: dry adiabat, $\Gamma \approx 8 \text{ K km}^{-1}$

Total atmospheric mass: ~ 90× Earth's; almost entirely CO₂.

Atmospheric Super-Rotation

Venus' atmosphere rotates much faster than the planet itself.

- Solid body: 243 days/rotation
- Cloud-top winds: $\sim 100 \text{ m s}^{-1}$, circling the planet in ~ 4 days
- Angular velocity ratio: $\sim 60\times$ the surface
- Driven by: thermal tides (day-night heating waves) + large-scale waves transporting angular momentum
- Persistent for the full mission record (~ 50 years)

Super-rotation is the textbook example of an atmospheric regime with no Earth analog and no easy first-principles derivation.

Venus' Greenhouse: T_{surf} from T_{eq}

Equilibrium temperature with $A = 0.77$:

$$T_{\text{eq}} = T_{\odot} \sqrt{R_{\odot} / (2a_V)} (1 - A)^{1/4} \approx 230 \text{ K}$$

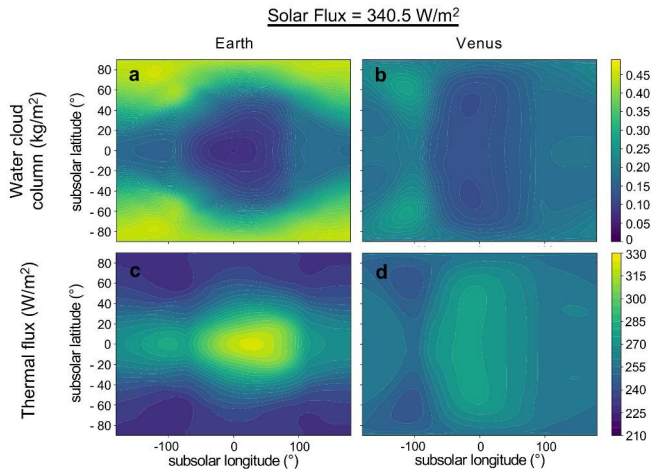
Actual surface temperature: 737 K.

Greenhouse effect: $\Delta T = +500 \text{ K}$.

- Earth: $\Delta T = +33 \text{ K}$
- Venus: $\Delta T = +500 \text{ K}$

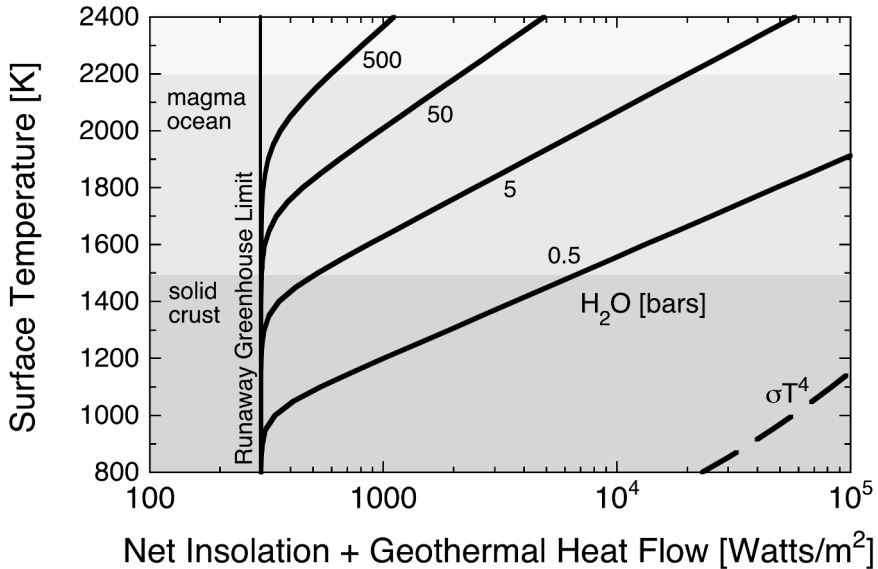
Venus is hot not because it is closer to the Sun (it absorbs less stellar flux than Earth!) but because of its CO_2 -dominated, 92-bar atmosphere.

Where the Night-Side Clouds Form



- Hot and steamy Earth (a, c) and Venus (b, d) at the same flux, 340.5 W/m^2
- Clouds gather on the night-side in both, although Venus turns 244 times slower: the pattern does not need slow rotation
- Night-side clouds trap thermal radiation, so the emission maps are anticorrelated with the cloud maps

Credit: Turbet et al. (2021), Fig. 2a-d

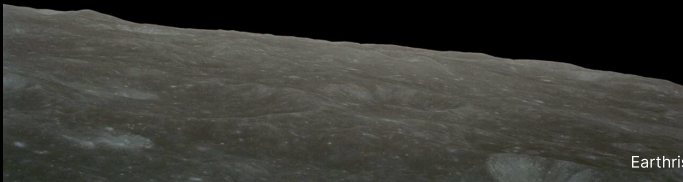


Reading the Runaway Threshold Figure

- Horizontal axis: net heat flux that must be radiated to space
- Vertical axis: surface temperature in steady state
- For low flux: cool, stable solution
- Vertical line on the right (Zahnle figure): runaway-greenhouse limit at $\sim 300 \text{ W m}^{-2}$; refined to $\sim 280 \text{ W m}^{-2}$ by Goldblatt et al. (2013) in the next slide
- Beyond the threshold: no steady state with surface liquid water; T_s **runs away** until water is fully vaporised

Maximum OLR sets the threshold. We derive it next, on the blackboard.

Break





Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Setup: Saturated Atmosphere, Greybody

Goal: show that a water-saturated atmosphere has a maximum OLR independent of surface temperature.

Assumptions:

- Atmosphere saturated in water vapour everywhere
- Single grey IR opacity κ , units $\text{m}^2 \text{kg}^{-1}$
- Hydrostatic equilibrium, surface gravity g
- Photosphere: where IR optical depth $\tau \sim 1$

Clausius–Clapeyron sets the water vapour partial pressure:

$$\frac{dp_{\text{sat}}}{dT} = \frac{L p_{\text{sat}}}{R_v T^2}$$

with $L \approx 2.5 \times 10^6 \text{ J kg}^{-1}$, $R_v = 461 \text{ J kg}^{-1} \text{ K}^{-1}$.

Step 1: The Photosphere Column Density

An IR optical depth of unity is reached at column density:

$$N_{\text{phot}} \sim \frac{1}{\kappa}$$

Hydrostatic equilibrium converts column density to pressure:

$$p_{\text{phot}} = g N_{\text{phot}} = \frac{g}{\kappa}$$

In a water-saturated atmosphere, the local water-vapour pressure equals saturation:

$$p_{\text{sat}}(T_{\text{phot}}) = \frac{g}{\kappa}$$

This single equation pins the photosphere temperature.

Step 2: Photosphere Temperature

Integrate Clausius–Clapeyron:

$$p_{\text{sat}}(T) \approx p_{\text{ref}} \exp \left[-\frac{L}{R_v} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right]$$

Set $p_{\text{sat}} = g/\kappa$ and solve for T :

$$T_{\text{phot}} \approx \frac{L/R_v}{\ln(p_{\text{ref}} \kappa/g) + L/(R_v T_{\text{ref}})}$$

Plug in $p_{\text{ref}} = 611 \text{ Pa}$, $T_{\text{ref}} = 273 \text{ K}$, $g = 10 \text{ m s}^{-2}$, $\kappa = 5 \times 10^{-2} \text{ m}^2 \text{ kg}^{-1}$:

$$T_{\text{phot}} \approx 260 \text{ K}$$

No surface temperature appears anywhere on the right-hand side.

Step 3: The OLR Asymptote

OLR is approximately blackbody emission at T_{phot} :

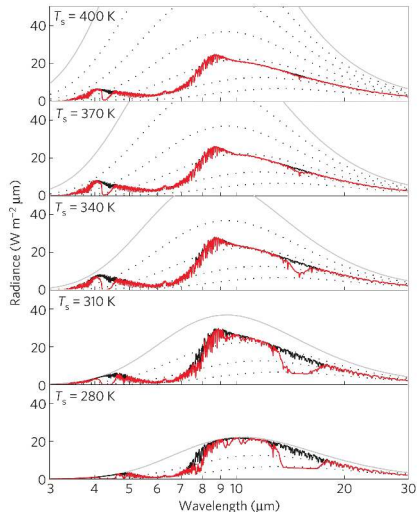
$$F_{\text{OLR}} \approx \sigma T_{\text{phot}}^4$$

$$F_{\text{OLR}}^{\text{max}} \approx 5.67 \times 10^{-8} \times (260)^4 \approx 260 \text{ W m}^{-2}$$

Refined line-by-line calculations: $F_{\text{OLR}}^{\text{max}} \approx 280\text{--}310 \text{ W m}^{-2}$ (Kasting 1988; Goldblatt et al. 2013).

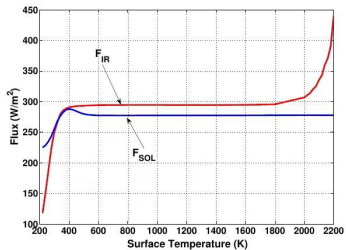
As T_s rises, the photosphere migrates upward but its temperature stays clamped: thermal flux saturates.

Numerical Verification: Goldblatt 2013

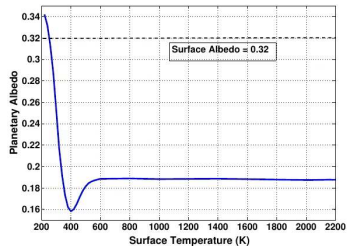


Credit: Goldblatt et al. (2013)

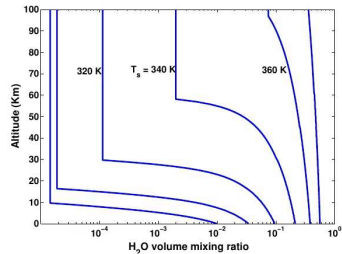
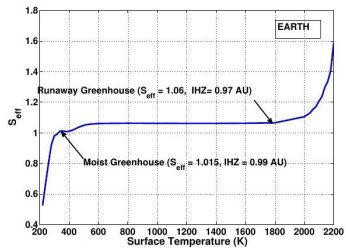
- Modelled OLR spectra at $T_s = 280, 310, 340, 370, 400$ K
- As T_s rises, the H_2O continuum closes the 8–14 μm window
- Integrated OLR saturates at $\sim 282 \text{ W m}^{-2}$ above $T_s \sim 340$ K
- Validates the Simpson–Nakajima sketch quantitatively



(a)



(b)

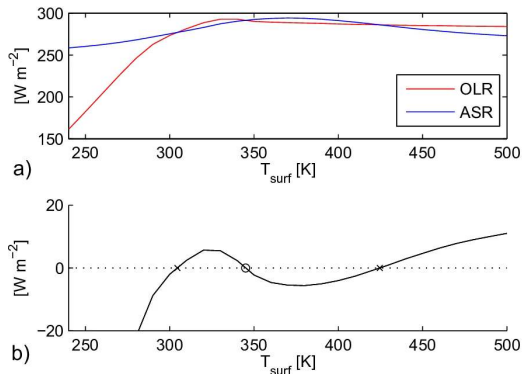


Reading the Kopparapu 2013 Panels

- Panel (a): F_{OLR} vs. T_s , asymptote at $\sim 291 \text{ W m}^{-2}$
- Panel (b): planetary albedo vs. T_s (surface albedo 0.32)
- Panel (c): S_{eff} vs. T_s with runaway ($S_{\text{eff}} = 1.06$) and moist-greenhouse (1.015) limits
- Panel (d): atmospheric H_2O volume mixing ratio vs. altitude
- Robust across reasonable choices of cloud + surface treatment

The Simpson–Nakajima limit is a thermodynamic feature, not a model artefact.

Multiple Equilibria and Hysteresis



Credit: Wordsworth & Pierrehumbert (2013)

- Three intersections: cool-stable, warm-unstable, hot-stable
- For a range of stellar fluxes, two stable solutions coexist
- Once on the hot branch, no return to cool without removing H_2O
- This is the source of the runaway hysteresis

Step 4: Habitable Zone Implications

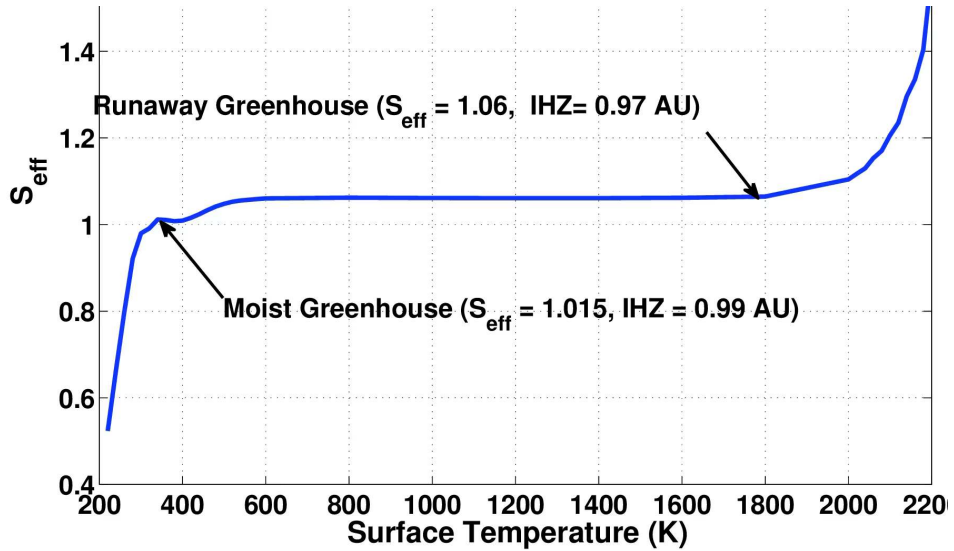
Compare to absorbed stellar flux at planetary orbit:

$$F_{\text{abs}} = \frac{S_{\odot}}{4a^2} (1 - A)$$

- Earth ($a = 1 \text{ AU}$, $A = 0.30$): $F_{\text{abs}} = 240 \text{ W m}^{-2}$, **below** $F_{\text{OLR}}^{\text{max}}$ by $\sim 40\text{--}70 \text{ W m}^{-2}$
- Venus ($a = 0.723 \text{ AU}$, Earth-like A): $F_{\text{abs}} \approx 460 \text{ W m}^{-2}$, **far above** $F_{\text{OLR}}^{\text{max}}$

Setting $F_{\text{abs}} = F_{\text{OLR}}^{\text{max}}$ gives the inner HZ edge:

$$a_{\text{IHZ}} \approx 0.95\text{--}0.99 \text{ AU} \quad (\text{present Sun})$$

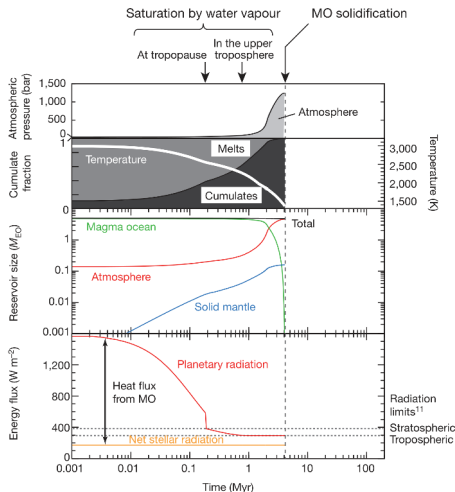


Reading the Kopparapu HZ Plot

- Surface temperature T_s on the x-axis, effective stellar flux S_{eff} on the y-axis
- **Runaway greenhouse** limit: $S_{\text{eff}} = 1.06$ ($a = 0.97$ AU around present Sun)
- **Moist greenhouse** limit: $S_{\text{eff}} = 1.015$ (0.99 AU)
- Both limits define the inner edge of the classical habitable zone
- Venus at $a = 0.723$ AU corresponds to $S_{\text{eff}} \approx 1.91$, well inside the limit

This figure is the canonical reference for the inner edge of the habitable zone.

Magma-Ocean Evolution: Type-I vs Type-II

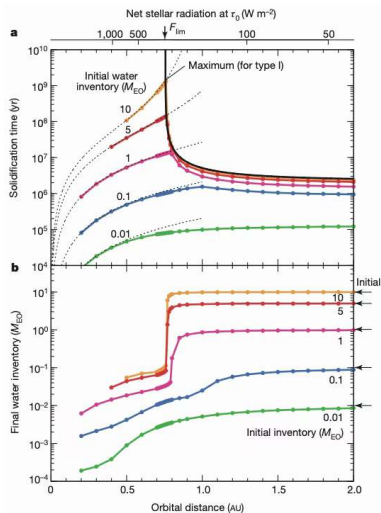


Credit: Hamano et al. (2013)

Type-I planets (outside critical orbit):

- Magma ocean (molten planetary surface layer) cools rapidly (< 10 Myr)
- Steam atmosphere condenses
- Liquid surface ocean forms
- Earth was Type-I

The Type-I / Type-II Phase Diagram

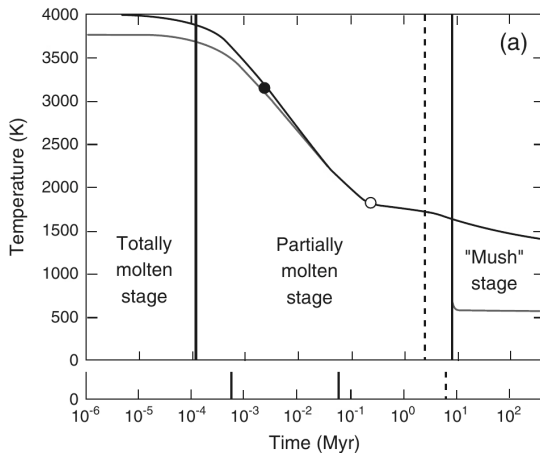


Credit: Hamano et al. (2013)

Critical orbital distance $a_{\text{cr}} \approx 0.77$ AU:

- Inside a_{cr} : **Type-II**, magma ocean persists > 100 Myr, water lost to space
- Outside a_{cr} : **Type-I**, ocean condenses, water retained
- Venus straddles or sits inside a_{cr}

Magma-Ocean Atmosphere Coupling



Credit: Lebrun et al. (2013)

Coupled magma-ocean + atmosphere model:

- Time evolution of T_{pot} and T_{surf} for a Venus-orbit planet
- Stages along the time axis: *totally molten* → *partially molten* → *mush*
- For $a \lesssim 0.66$ AU (separate panel, not shown): solidification is indefinitely delayed
- Establishes the timing baseline for Venus' steam-atmosphere phase

Why the Simpson-Nakajima Limit is Robust

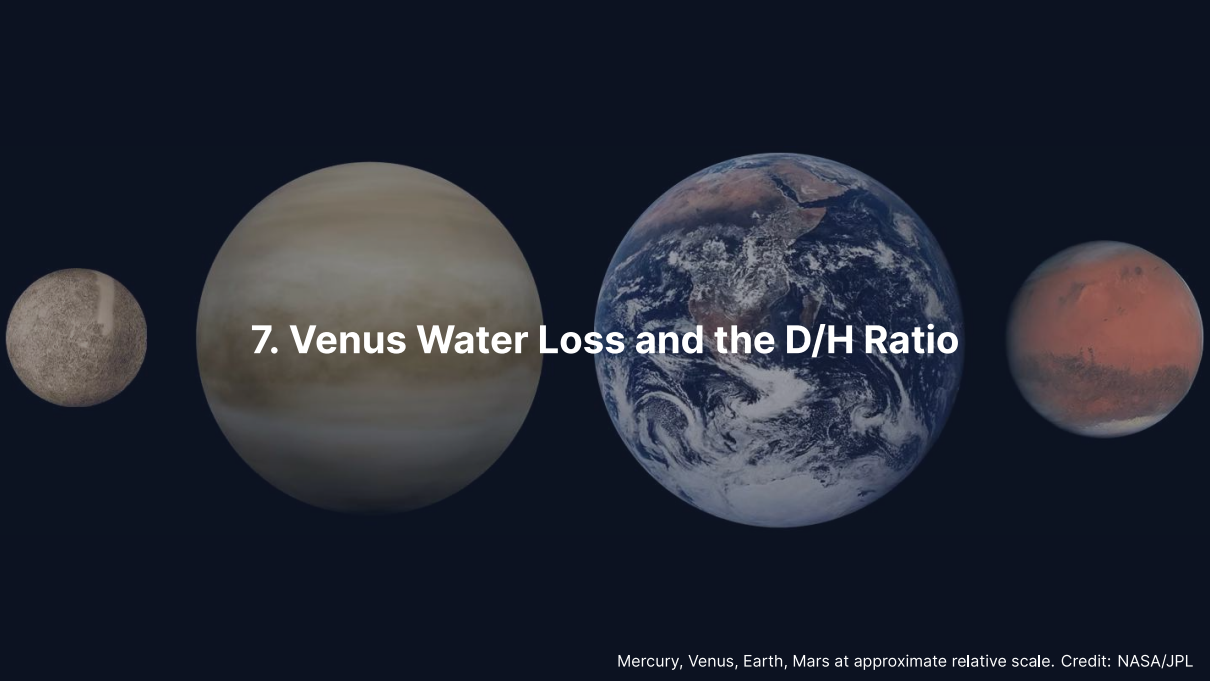
The limit depends only on:

- Water phase diagram (Clausius–Clapeyron)
- Elementary radiative transfer ($\tau \sim 1$ photosphere)

Insensitive to:

- Atmospheric composition (so long as water is abundant)
- Surface temperature (by construction)
- Detailed cloud microphysics (corrections at the 5–10% level)

Once $F_{\text{abs}} > F_{\text{OLR}}^{\text{max}}$, no steady state with surface liquid water. The transition is one-way: only photolysis + H escape returns the planet to a dry state, irreversibly.

A horizontal arrangement of four celestial bodies against a dark blue background. From left to right: Mercury, a small greyish-brown sphere; Venus, a large pale yellowish-tan sphere; Earth, a large blue and white sphere showing continents and clouds; and Mars, a medium-sized reddish-orange sphere. The text '7. Venus Water Loss and the D/H Ratio' is centered over the Venus and Earth spheres.

7. Venus Water Loss and the D/H Ratio

Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

When Did Venus Lose Its Water?

Two end-member scenarios, both physically plausible:

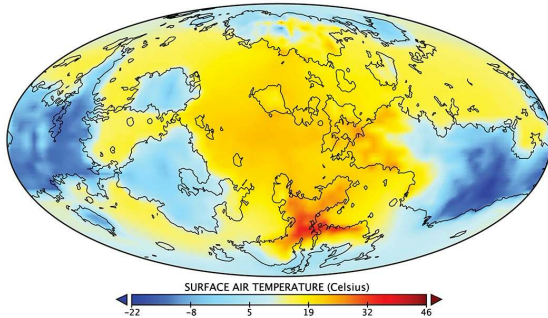
- **Early loss:** runaway from the magma-ocean stage; never had condensed oceans (Hamano Type-II)
- **Late loss:** temperate Earth-like surface for ~ 1 Gyr, runaway triggered by the Sun's increasing main-sequence luminosity (Way et al. 2016)

Distinguishing observation needed: surface mineralogy + atmospheric isotopes.

None of the present datasets is conclusive.

Was Venus Ever Habitable? The Late-Loss Scenario

(a) Paleo Venus Surface Air Temperature (2.9Gya solar spectrum)



Credit: Way et al. (2016)

- GCM snapshot at 2.9 Gya: surface T ranges from -22°C polar to $+46^{\circ}\text{C}$ equatorial
- Cloud feedback from slow rotation reflects flux back to space
- Across the Way et al. (2016) ensemble, global mean $T < 290\text{ K}$ can persist for $\sim 1\text{ Gyr}$
- A habitable Venus is dynamically possible, not required

The Loss Mechanism: Photolysis + Escape

Sequential loss of water in the upper atmosphere:

1. $\text{H}_2\text{O} + h\nu \rightarrow \text{H} + \text{OH}$ (UV photolysis)
2. H escapes via Jeans + non-thermal channels (light, fast)
3. O recombines, slowly oxidises crust or escapes via solar-wind pickup

Mass balance: ~1 ocean equivalent of water can be lost over Gyr through this channel.

Without a magnetic field, the solar wind directly accesses the upper atmosphere $\Rightarrow$ enhanced O^+ pickup.

The D/H Ratio: Smoking-Gun Evidence

Measured atmospheric D/H ratios:

Body	D/H ($\times 10^{-4}$)
Protosolar (H_2)	~ 0.20
Earth (oceans)	1.56
Comets (Halley, 67P)	1.5–5
Venus	~ 240
Mars	~ 8

Source: Donahue et al. (1982); de Bergh et al. (1991)

- Venus' D/H is $\sim 150\times$ **Earth's**
- Lighter 1H escapes more easily than heavier 2H
- D enrichment $\Rightarrow$ massive past H loss
- Initial water $\geq 1\%$ of Earth's ocean mass

Bounding the Initial Water Inventory

Rayleigh fractionation in escape: residual water becomes D-enriched as H is lost.

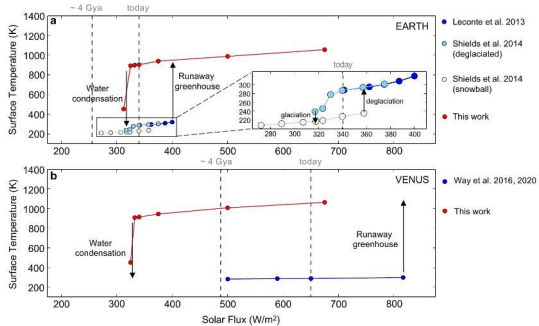
Two limiting fractionation regimes:

- **Diffusion-limited:** $\alpha \approx m_{\text{H}}/m_{\text{D}} \approx 0.5 \Rightarrow f \approx 150^{-2} \approx 4 \times 10^{-5}$
- **Bare-Jeans:** $\alpha \approx \sqrt{m_{\text{H}}/m_{\text{D}}} \approx 0.71 \Rightarrow f \approx 150^{-3.4} \approx 4 \times 10^{-8}$

Multiplying $1/f$ by the present water column (~2 cm GEL) gives initial inventories of **several hundred metres** (diffusion-limited) up to unphysically large values in the bare-Jeans limit.

Either way: $\geq 100\times$ today's water (consistent with the previous slide's $\geq 1\%$ Earth-ocean lower bound).

Bistability and Hysteresis Loop



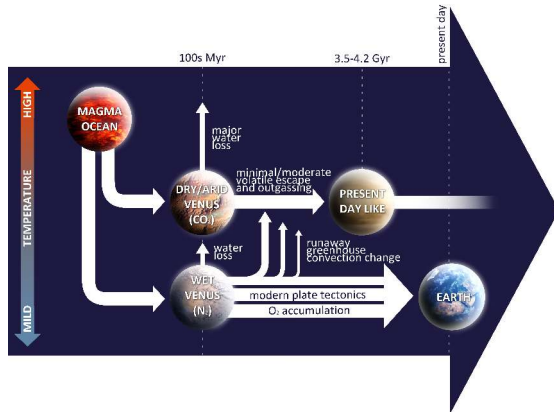
Credit: Turbet et al. (2021), Fig. 4

For some stellar fluxes, two stable states coexist:

- Wet: ocean + thin atmosphere
- Dry: hot, thick CO_2 atmosphere

Crossing the threshold is one-way. The dry state has no thermodynamic path back.

Dry-Venus vs Wet-Venus Pathways



Credit: Gillmann et al. (2022)

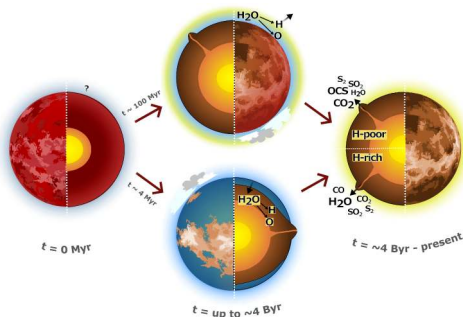
Coupled-model branching:

- Initial water inventory + outgassing rate determines branch
- Wet branch: like Earth, ocean condenses
- Dry branch: like Venus today
- Initial conditions decide everything

Constraining the Trajectory

Two end-member pathways for Venus' water history:

- **Top branch (dry):** no surface ocean; H lost early; mantle dry today
- **Bottom branch (wet):** ocean condenses, late loss; mantle still wet today
- Constantinou et al. (2024) argue current photochemistry favours the dry/early-loss branch (volcanic $\text{H}_2\text{O} \lesssim 6\%$)
- DAVINCI noble-gas measurements will discriminate



Credit: Constantinou et al. (2024)

A horizontal arrangement of four planets against a dark blue background. From left to right: Mercury is a small, grey, cratered sphere. Venus is a large, pale yellowish-brown sphere with faint horizontal cloud bands. Earth is a large sphere showing blue oceans, white clouds, and brown/green landmasses. Mars is a medium-sized, reddish-brown sphere with darker patches. The text "8. Why Did Earth and Venus Diverge?" is centered over the Venus and Earth planets.

8. Why Did Earth and Venus Diverge?

Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

Carbonate–Silicate Cycle: Failure Mode

On Earth, the carbon cycle keeps CO_2 in the atmosphere ~ 280–1000 ppm via:

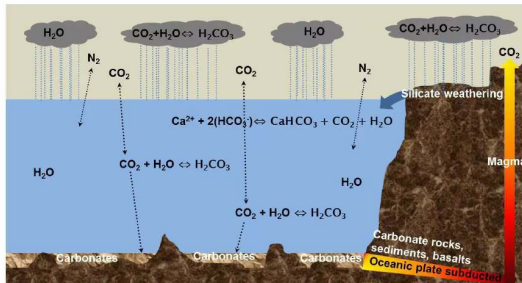
- Volcanic outgassing (source)
- Silicate weathering on continents (sink, T -dependent)
- Carbonate subduction + decarbonation (return loop)

On Venus this cycle has **no functional sink**:

- No liquid water $\Rightarrow$ no aqueous weathering
- No subduction $\Rightarrow$ no return of crustal carbonate to the mantle
- Volcanic outgassing accumulates indefinitely

Result: CO_2 partial pressure builds to the 90-bar ambient observed today.

The Earth Carbonate-Silicate Schematic



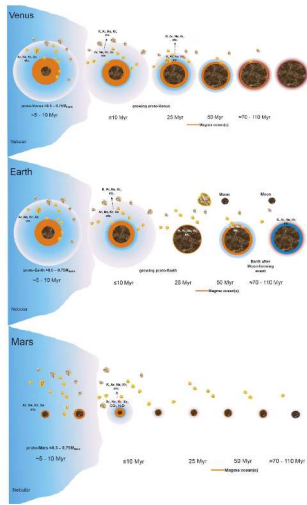
Credit: Lammer et al. (2018)

Earth's closed carbon loop:

- Volcanic outgassing (mantle to atmosphere)
- Silicate weathering on continents (atmosphere to ocean, T -dependent sink)
- Carbonate sedimentation + subduction (ocean to mantle return leg)

Venus today has no liquid-water sink and no subduction return: both halves of the loop are broken (next slide).

Initial Volatile Inventories



Credit: Lammer et al. (2018)

- Comparative accretion of Venus, Earth, Mars
- Initial water + carbon inventories overlap substantially
- Divergence is not from initial conditions alone
- Early evolution decides which branch each planet takes

Why Did the Twins Diverge?

No single cause; several contributing factors:

1. **Distance from the Sun** ($1.91\times$ Earth's flux) $\Rightarrow$ inside the runaway-greenhouse limit
2. **No active dynamo** $\Rightarrow$ no shielding from solar wind
3. **Slow rotation** $\Rightarrow$ less Coriolis-driven cloud organisation
4. **Possible early magma-ocean phase**: Hamano Type-II, volatiles never fully condensed
5. **Stagnant lid tectonics** $\Rightarrow$ no carbon-cycle return loop

All five may have contributed; their relative weight is an active research question.

Implications for Habitability

Venus matters far beyond Venus:

- Defines the inner edge of the classical habitable zone for any star
- Provides a **negative example** of what happens when key feedbacks fail
- Stagnant-lid + runaway combination may be the default for rocky planets, with Earth as the lucky exception
- Tidally-locked rocky planets near M-dwarfs are pre-conditioned to Venus-like outcomes

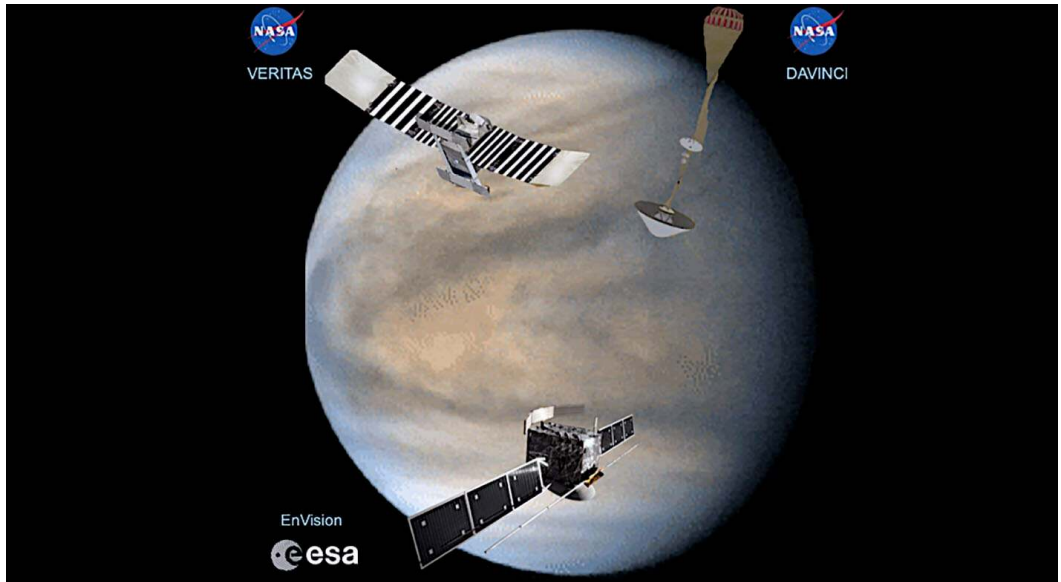
Venus is the calibration target for understanding which combinations of orbital, geophysical, and tectonic parameters allow long-term liquid water.



9. Recent Advances and Future Missions



VERITAS, DAVINCI and EnVision over an Akatsuki UV view of Venus. Credit: Widemann et al. (2023)



DAVINCI (NASA), VERITAS (NASA), EnVision (ESA). Widemann et al. (2023)

Three Missions to Venus, This Decade

- **DAVINCI** (NASA, ~ 2030): atmospheric descent probe + flyby imaging
- **VERITAS** (NASA, ~ 2031): radar interferometry, surface composition mapping
- **EnVision** (ESA, ~ 2031): radar + subsurface sounder + atmospheric chemistry

Combined, they will measure: noble-gas isotopes (DAVINCI), high-resolution topography (VERITAS), subsurface structure + tidal Love number k_2 , a measure of tidal deformation (EnVision).

After a ~ 30-year gap in dedicated Venus missions, three at once.

DAVINCI: A Descending Atmospheric Laboratory

- Launch: nominally 2029–2030
- Hour-long parachute descent through the full atmosphere
- Mass spectrometer: noble gases + isotopes (Ar, Kr, Xe ratios)
- Tunable laser spectrometer: D/H, $^{18}\text{O}/^{16}\text{O}$ at multiple altitudes
- Imaging: descent images of tessera highlands at < 100 m resolution

Will measure noble-gas isotopes that distinguish early-loss vs. late-loss water scenarios for the first time.

VERITAS: Radar at 30 m Resolution

- Synthetic aperture radar imagery: ~ 30 m/px (Magellan SAR: ~ 100–300 m/px)
- InSAR-derived topography: ~ 250 m posting, ~ 5 m vertical (Magellan altimetry: ~ 10–20 km horizontal)
- InSAR change detection: cm-scale surface deformation between flybys
- Near-IR emissivity: surface mineralogy through the atmosphere
- Gravity science: high-resolution gravity map, C/MR^2 measurement

Will measure C/MR^2 for the first time, settling the core-size question.

EnVision: Surface, Subsurface, Atmosphere

- ESA M5 mission, launch ~ 2031
- Synthetic aperture radar (VenSAR): selected high-resolution targets
- **Subsurface sounder**: layered structure of plains and tessera to ~ 1 km depth
- Spectrometers: mineralogy + atmospheric chemistry
- Radio science: gravity tides, k_2 Love number $\Rightarrow$ core liquidity

k_2 measurement will distinguish a liquid from solid core, addressing the missing-dynamo question.

Open Questions for the 2030s

1. Did Venus ever have surface oceans? For how long?
2. What is the present rate of volcanism?
3. Is the core liquid? Why no dynamo?
4. Does the atmosphere host disequilibrium chemistry of biological origin?
5. Does episodic-overtake tectonics operate, and on what timescale?

Venus is moving from a puzzle with no data to a puzzle with the right data on the way.

Summary

1. Earth and Venus have nearly identical bulk parameters
2. Their evolution diverged by ~ 500 K in surface temperature, $90\times$ in atmospheric mass, all surface water vs. none
3. Earth: plate tectonics + magnetic dynamo + biosphere + carbon cycle, all mutually reinforcing
4. Venus: stagnant lid + no dynamo + runaway greenhouse, all mutually reinforcing
5. The Simpson–Nakajima limit ($\sim 280 \text{ W m}^{-2}$) is the thermodynamic boundary defining the inner edge of the habitable zone
6. Venus' D/H ratio of $\sim 150\times$ Earth's is direct evidence of past water loss
7. DAVINCI, VERITAS, and EnVision will deliver the data to settle when, why, and how the divergence happened



Questions?

Next: Lecture 10. Rocky Planets II: Mercury & Mars

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience

